

Begum Rokeya Univesity,Rangpur

Department of Statistics

STAT-2205: Python Programming Project

Flight Data Analysis

ID: 12310066

```
pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Load dataset
```

```
flight =  
pd.read_csv("C:/Users/USER/Downloads/  
R & Python/flight_data.csv")
```

```
# Random sample using student ID
```

```
flight_sample = flight.sample(n=10000,  
random_state=66)
```

```
## Q.1: What do coach ticket prices  
look like? What are the high and  
low values? What would be  
considered the average? Does $500  
seem like a good price for a coach  
ticket?"
```

```
Solution
```

```
#Summary statistics
```

```
print(flight_sample['coach_price'].describe  
())
```

```
# Mean
```

```
mean_price =  
flight_sample['coach_price'].mean()  
  
print("Average Coach Price:",  
mean_price)
```

```
# Histogram
```

```
plt.hist(flight_sample['coach_price'],  
bins=30)
```

```
plt.xlabel("Coach Ticket Price")
```

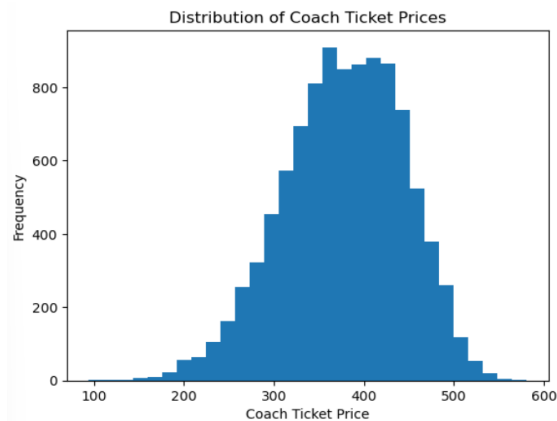
```
plt.ylabel("Frequency")
```

```
plt.title("Distribution of Coach Ticket  
Prices")
```

```
plt.show()
```

Output

```
count    10000.000000  
mean       377.572120  
std        67.287038  
min        94.535000  
25%       333.180000  
50%       380.842500  
75%       427.177500  
max       580.850000  
Name: coach_price, dtype: float64  
Average Coach Price: 377.57212000000000  
4
```



##Q2: What are the coach prices for flights that are 8 hours long? Does a \$500 ticket seem more reasonable for these flights?

Solution

8-hour flights

```
flight_8hr =
flight_sample[flight_sample['hours'] == 8]
```

Summary statistics

```
print(flight_8hr['coach_price'].describe())
```

Output

```
count    212.000000
mean     438.763821
std       57.962725
min      242.600000
25%      405.296250
50%      446.252500
75%      478.943750
max      561.750000
Name: coach_price, dtype: float64
```

##Q3: What does the distribution of flight delays look like? Are there any outliers?

Solution

```
import matplotlib.pyplot as plt
```

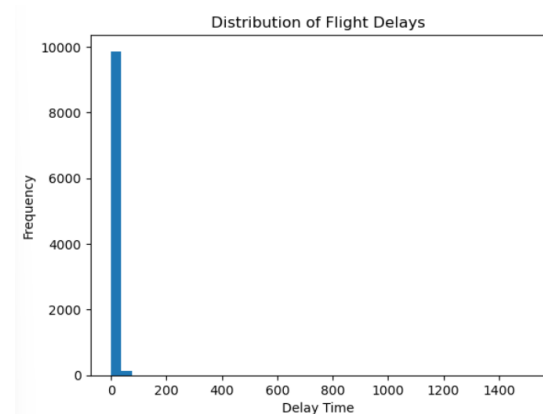
Histogram of delays

```
plt.hist(flight_sample['delay'], bins=40)
plt.xlabel("Delay Time")
plt.ylabel("Frequency")
plt.title("Distribution of Flight Delays")
plt.show()
```

Summary statistics

```
print(flight_sample['delay'].describe())
```

Output



```
count    10000.000000
mean       13.139800
std        39.697635
min         0.000000
25%         9.000000
50%        10.000000
75%        13.000000
max       1497.000000
Name: delay, dtype: float64
```

##Q4: How are coach ticket prices related to flight distance (miles)?

Solution

```
import seaborn as sns
import matplotlib.pyplot as plt
```

Scatter plot

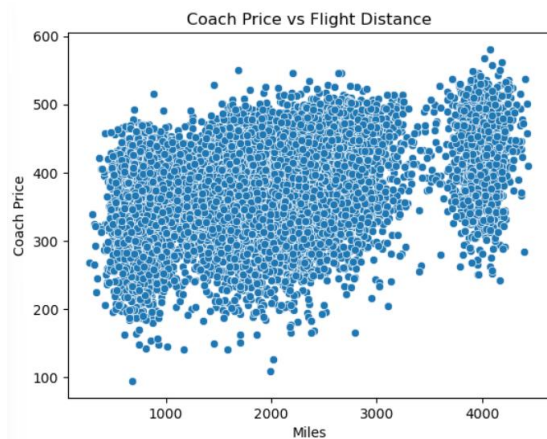
```
sns.scatterplot(
    x='miles',
    y='coach_price',
    data=flight_sample
)
```

```
plt.xlabel("Miles")
plt.ylabel("Coach Price")
plt.title("Coach Price vs Flight Distance")
plt.show()
```

Correlation

```
print(flight_sample[['miles','coach_price']].corr())
```

Output



```
miles coach_price
miles      1.000000    0.372265
coach_price 0.372265    1.000000
```

##Q5: How are flight hours related to coach ticket prices?

Solution

```
import seaborn as sns
import matplotlib.pyplot as plt
```

Scatter plot

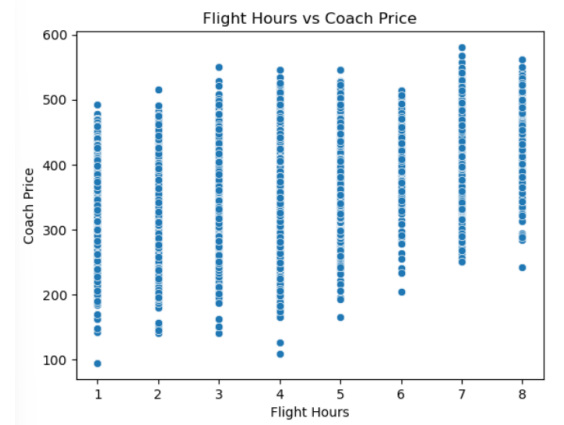
```
sns.scatterplot(
    x='hours',
    y='coach_price',
    data=flight_sample
)
```

```
plt.xlabel("Flight Hours")
plt.ylabel("Coach Price")
plt.title("Flight Hours vs Coach Price")
plt.show()
```

Correlation

```
print(flight_sample[['hours','coach_price']].corr())
```

Output



```
hours coach_price
hours      1.000000    0.366395
coach_price 0.366395    1.000000
```

##Q6: How are coach ticket prices related to first-class ticket prices?

Solution

```
import seaborn as sns
import matplotlib.pyplot as plt
```

Scatter plot

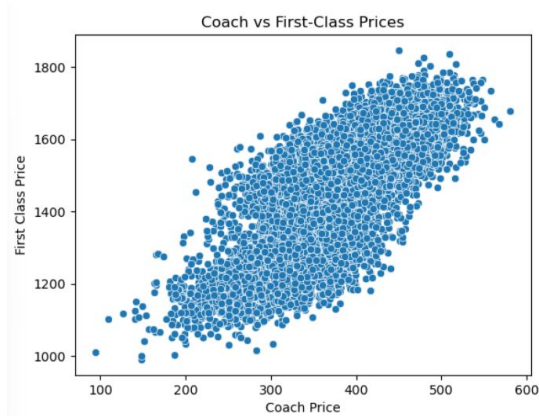
```
sns.scatterplot(
    x='coach_price',
    y='firstclass_price',
    data=flight_sample
)
```

```
plt.xlabel("Coach Price")
plt.ylabel("First Class Price")
plt.title("Coach vs First-Class Prices")
plt.show()
```

Correlation

```
print(flight_sample[['coach_price',
                    'firstclass_price']].corr())
```

Output



```

coach_price firstclass_price
coach_price    1.000000    0.757325
firstclass_price 0.757325    1.000000

```

##Q7: How do in-flight features affect coach ticket prices?

Solution

Average coach prices by features

```

print(flight_sample.groupby(
    'inflight_meal')['coach_price'].mean())

```

```

print(flight_sample.groupby(

```

```

'inflight_entertainment')['coach_price'].me
an())

```

```

print(flight_sample.groupby(
    'inflight_wifi')['coach_price'].mean())

```

Output

```

inflight_meal
No    371.829576
Yes   391.282637
Name: coach_price, dtype: float64
inflight_entertainment
No    322.052406
Yes   391.209997
Name: coach_price, dtype: float64
inflight_wifi
No    314.918785

```

Yes 384.310008

Name: coach_price, dtype: float64

##Q8: How do weekend flights compare to weekday flights in terms of ticket prices?

Solution

```

import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

```

Average ticket prices

```

weekend_price = flight_sample.groupby('
weekend')[
    ['coach_price', 'firstclass_price']
].mean()

```

```

print(weekend_price)

```

Bar plot

```

weekend_price.plot(kind='bar')

```

```

plt.title("Weekend vs Weekday Ticket Prices")
plt.ylabel("Average Price")
plt.xticks(rotation=0)
plt.show()

```

Output

```

coach_price firstclass_price
weekend
No    323.047448    1258.846526
Yes    406.088374    1559.762531

```



##Q9: How do redeye flights compare to non-redeye flights in terms of coach ticket prices?

Solution

```
import seaborn as sns
import matplotlib.pyplot as plt

# Average coach prices
redeye_price = flight_sample.groupby('redeye')[
    'coach_price'
].mean()

print(redeye_price)

# Bar plot
redeye_price.plot(kind='bar')

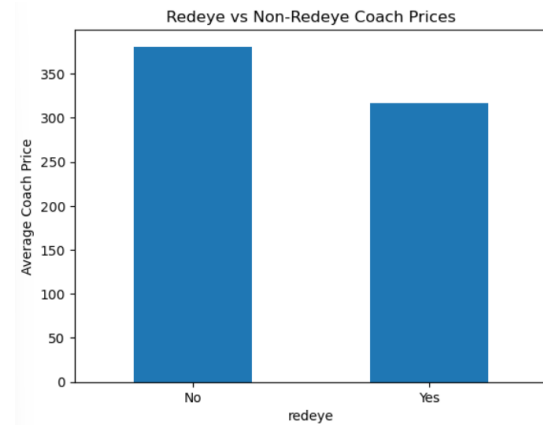
plt.title("Redeye vs Non-Redeye Coach Prices")
plt.ylabel("Average Coach Price")
plt.xticks(rotation=0)
plt.show()
```

Output

```
redeye
No    380.715426
```

Yes 316.435143

Name: coach_price, dtype: float64



##Q10: Perform the analysis for appropriate variables

Solution

##(1)Summary

```
flight_sample.describe()
```

Output

	Miles	passengers	delay	coach_price	firstclass_price	hours
count	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000	10000.000000
mean	2011.737400	207.653500	13.139800	377.572120	1456.427975	3.635600
std	955.070753	12.714905	39.697635	67.287038	162.780237	1.760092
min	279.000000	143.000000	0.000000	94.535000	990.010000	1.000000
25%	1320.000000	204.000000	9.000000	333.180000	1304.675000	2.000000
50%	1982.500000	209.000000	10.000000	380.842500	1503.420000	4.000000

75%	2475.000000	215.000000	13.000000	427.177500	1584.047500	4.000000
max	4438.000000	236.000000	1497.000000	580.850000	1846.780000	8.000000

#10(2) Visualization

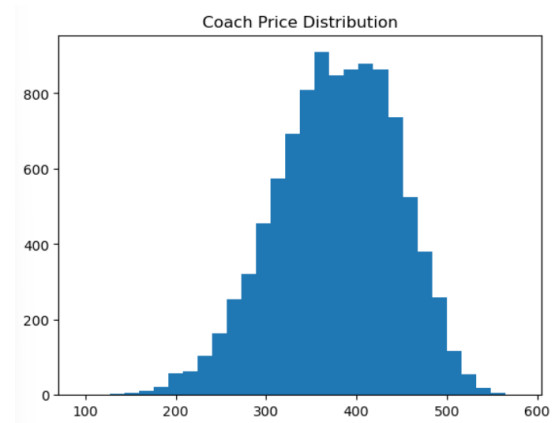
```
import matplotlib.pyplot as plt
```

Histogram

```
plt.hist(flight_sample['coach_price'],
bins=30)

plt.title("Coach Price Distribution")

plt.show()
```

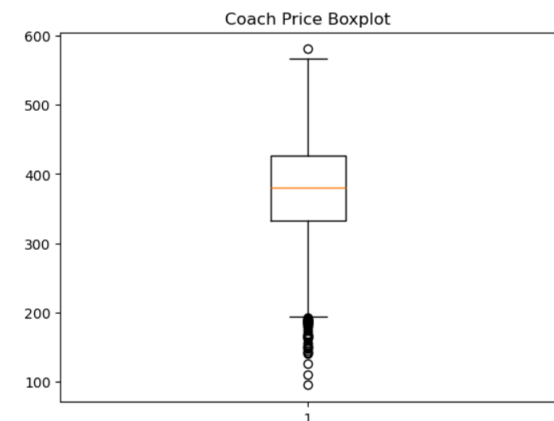


Boxplot

```
plt.boxplot(flight_sample['coach_price'])

plt.title("Coach Price Boxplot")

plt.show()
```

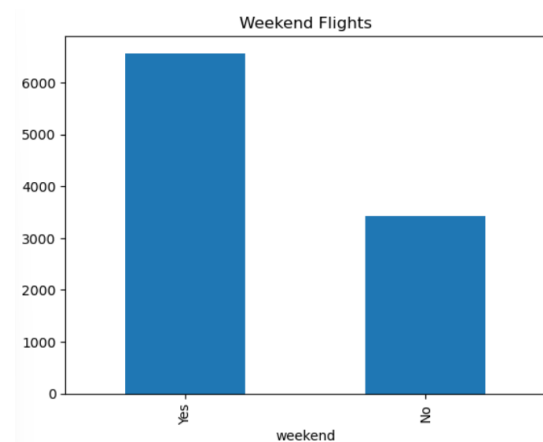


Bar chart

```
flight_sample['weekend'].value_counts().plot(kind='bar')

plt.title("Weekend Flights")

plt.show()
```



Output

##10(c) Hypothesis Testing (t-test)

```
from scipy import stats
```

```
weekend_yes = flight_sample[  
    flight_sample['weekend']=="Yes"  
]['coach_price']
```

```
weekend_no = flight_sample[  
    flight_sample['weekend']=="No"  
]['coach_price']  
stats.ttest_ind(weekend_yes, weekend_no)
```

output

```
TtestResult(statistic=np.float64(72.319668  
1400005), pvalue=np.float64(0.0), df=np.f  
loat64(9998.0))
```

##10(d) Correlation Analysis

```
flight_sample[  
    ['coach_price',  
     'miles',  
     'hours',  
     'delay']  
].corr()
```

Output

	coach_price	miles	hours	delay
coach_price	1.000000	0.372265	0.366395	0.018023
miles	0.372265	1.000000	0.985873	0.005649
hours	0.366395	0.985873	1.000000	0.008739
delay	0.018023	0.005649	0.008739	1.000000

##10(e) Regression Analysis

```
from sklearn.linear_model import LinearR  
egression
```

```
X = flight_sample[  
    ['miles','hours','delay']  
]
```

```
y = flight_sample['coach_price']
```

```
model = LinearRegression()  
model.fit(X,y)
```

```
print(model.coef_)
```

Output

```
[ 0.02785972 -0.90216435  0.02711225]
```

##10(f) Logistic Regression

```
from sklearn.linear_model import  
LogisticRegression
```

```
flight_sample['redeye_binary'] = (  
    flight_sample['redeye']=="Yes"  
).astype(int)
```

```
X = flight_sample[  
    ['hours','coach_price']  
]
```

```
y = flight_sample['redeye_binary']
```

```
log_model = LogisticRegression()  
log_model.fit(X,y)
```

```
print(log_model.score(X,y))
```

Output

```
0.9508
```

```
from sklearn.linear_model import LinearR  
egression
```

```
X = flight_sample[  
    ['miles',
```

```
'hours',  
'delay']  
]
```

```
y = flight_sample['coach_price']
```

```
model = LinearRegression()
```

```
model.fit(X, y)
```

```
print(model.coef_)
```

```
print(model.intercept_)
```

Output

```
[ 0.02785972 -0.90216435  0.02711225]
```

```
324.44933798732814
```

```
import
```